

Project Title: Nightingale, A Hospital Management App

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Phase 1:

Description:

Nightingale, The Hospital Management Android Application, is a **mobile-based software system** designed to automate and digitize the management of hospital operations. The application will be developed using **Kotlin** on the Android platform to provide a secure, efficient, and user-friendly solution for managing hospital services.

The system will support three types of users: **Admin**, **Doctor**, and **Patient**. Each user will have specific access and functionalities according to their roles.

The Admin will be responsible for managing hospital resources such as bed allotment, operation theatre booking, diagnostic test scheduling, and managing doctors' schedules. This will help ensure efficient utilization of hospital facilities and improve operational management.

The Doctor will be able to set available appointment times, view patient medical history, provide prescriptions, and monitor patient treatment. Doctors will also be able to check the availability of medicines in the hospital pharmacy and review nursing routines.

The Patient will be able to book appointments with doctors, access prescriptions, view diagnostic test results, and check their medical history. This will allow patients to easily access their medical information and healthcare services.

The application will reduce manual paperwork, minimize errors, improve communication between hospital staff and patients, and increase overall efficiency in hospital

management.

Objective:

General Objective: To design and develop a secure, efficient, and user-friendly Android-based Hospital Management System that enables digital management of hospital operations and healthcare services.

Specific Objectives:

1. To develop an Admin module to manage hospital resources such as bed allotment, operation theatre booking, diagnostic test scheduling, and doctor scheduling.
2. To develop a Doctor module to allow doctors to manage appointment availability, view patient medical history, provide prescriptions, and monitor patient treatment.
3. To develop a Patient module to enable patients to book appointments, view prescriptions, access test results, and maintain medical history.
4. To create a centralized and secure database system for storing hospital and patient information.
5. To reduce paperwork and manual errors by automating hospital management processes.
6. To improve communication between Admin, Doctor, and Patient.
7. To provide convenient and easy access to hospital services through a mobile application.
8. To ensure the security, privacy, and integrity of medical data.

Phase 2:

Software Requirement Specification(SRS)

For

Nightingale,

A Hospital Management App

Version 1.0

Prepared By - Group-8

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1.Introduction

1.1 Project Title

Nightingale, A Hospital Management App

1.2 Purpose

The purpose of the ***Nightingale ,A Hospital Management App*** is to make hospital services easier, faster, and more organized for everyone involved-patients, doctors, and administrators.

In many hospitals, tasks like booking appointments, managing patient records, writing prescriptions, and handling admissions are still done manually or through scattered systems. This often leads to delays, errors, and miscommunication. Our project aims to solve these problems by bringing everything into one simple mobile application.

The main goal is to reduce paperwork, improve communication, and ensure that healthcare services become more accessible and reliable. By digitizing these processes, the system helps save time, minimize errors, and create a smoother experience for both healthcare providers and patients.

This project is about transforming traditional hospital management into a smart, connected, and user-friendly digital system.

1.3 Intended Audience

Software Developers

UI/UX Designers

Project Supervisors

Stakeholders

1.4 Scope

Nightingale is an Android-based mobile application developed using Kotlin that aims to digitize and automate hospital management processes. The system will support three

primary user roles:

- ❖ Admin
- ❖ Doctor
- ❖ Patient

The application will streamline hospital operations, improve communication, and ensure efficient healthcare service delivery.

2. Overall Description

2.1 User Roles

User Type	Description
Admin	Manages hospital resources, schedules, and overall system control
Doctor	Provides medical services, prescriptions, and monitors patients
Patient	Books appointments, views medical data, and interacts with doctors

2.2 Operating Environment

The *Nightingale Hospital Management App* operates in a mobile and cloud-based environment to ensure smooth performance and accessibility.

Client-Side

Platform: Android
Language: Kotlin
OS Requirement: Android 8.0 or above
Runs on smartphones with internet access

Server-Side

Backend: Firebase (or similar cloud service)
Provides authentication, database, and notification services

Database

Cloud-based (NoSQL) database
Stores patient records, appointments, prescriptions, and other hospital data securely

Network

Requires stable internet connection
Supports real-time data synchronization

User Environment

Accessible by Admins, Doctors, and Patients
Secure login with a simple and user-friendly interface

2.3 Assumptions

- The system will run on Android devices
- Internet connectivity is required
- Secure authentication is mandatory for all users
- Data privacy regulations must be followed

3. Functional Requirements(FR)

FR ID	Module	Functional Requirements
FR-1	User Authentication	- Secure registration and login - Role-based access: Admin, Doctor, Patient - Password encryption and secure session handling
FR-2	Admin Module	- Manage hospital beds, operation theatres, and diagnostic tests - Add/update/remove doctor profiles and manage schedules - View system activities and generate reports
FR-3	Doctor Module	- Set and manage appointment slots - Access patient history and monitor treatments - Create/share prescriptions digitally - Check medicine availability
FR-4	Patient Module	- Search, book, and cancel appointments - View prescriptions, diagnostic results, and maintain medical history - Receive notifications and updates from doctors/admin

FR ID	Module	Functional Requirements
FR-5	Database Management	- Centralized storage of patients, doctors, appointments, and prescriptions - Ensure data consistency and integrity

4.Non-Functional Requirements(NFR)

NFR ID	Category	Description
NFR-1	System Performance	Fast response time (< 3 seconds) and efficient handling of concurrent users.
NFR-2	Information Security	Data encryption (at rest and in transit), robust authentication and authorization, protection against unauthorized access.
NFR-3	User Experience (UX)	Intuitive and user-friendly interface, seamless navigation, minimal learning curve.
NFR-4	System Reliability	High availability (24/7) with minimal downtime and fault tolerance.

NFR ID	Category	Description
NFR-5	Scalability & Extensibility	Capability to handle increasing numbers of users and expanding data volume.
NFR-6	Maintainability & Modularity	Modular architecture to facilitate easy updates, enhancements, and bug fixes.

5. External Interfaces

User Interface(UI)

- Web and mobile interfaces for patients to search doctors, select departments, book appointments, and view statuses.
- Doctor dashboard to accept, reject, or manage appointments.

APIs

- RESTful APIs for appointment creation, status updates, and user authentication.
- Integration with notification services (email/SMS) for reminders.

Hardware

- Standard client devices (PC, smartphones, tablets).
- Servers to host backend services and databases.

Database

- Relational database(eg. SQL) to store user, doctor, department, and appointment data.
- Support for secure data retrieval and updates.

Other Interfaces

- Payment gateways (if online payments are implemented).

6. Constraints & Assumptions

- Real-time appointment updates require internet connectivity; offline support is not included.
- User authentication (e.g., login, OTP) assumes valid credentials are provided.
- All patient and doctor data must be stored on university-approved or secure

servers.

- Notification delivery (email/SMS) depends on third-party services availability. • Appointment conflicts are prevented by system logic, but manual overrides by staff may cause inconsistencies.
- User consent is assumed for storing personal and medical-related data (compliant with GDPR/local data laws).

7. Appendices

- **Firebase:** A cloud-based platform used for the app's backend, authentication, and real-time database.
- **RESTful API:** An application programming interface that allows the Android app to communicate with the server.
- **OTP:** One-Time Password used for secure user authentication. • **GDPR:** General Data Protection Regulation; the law governing data privacy and user consent.
- **Kotlin:** The primary programming language used to build the Android application

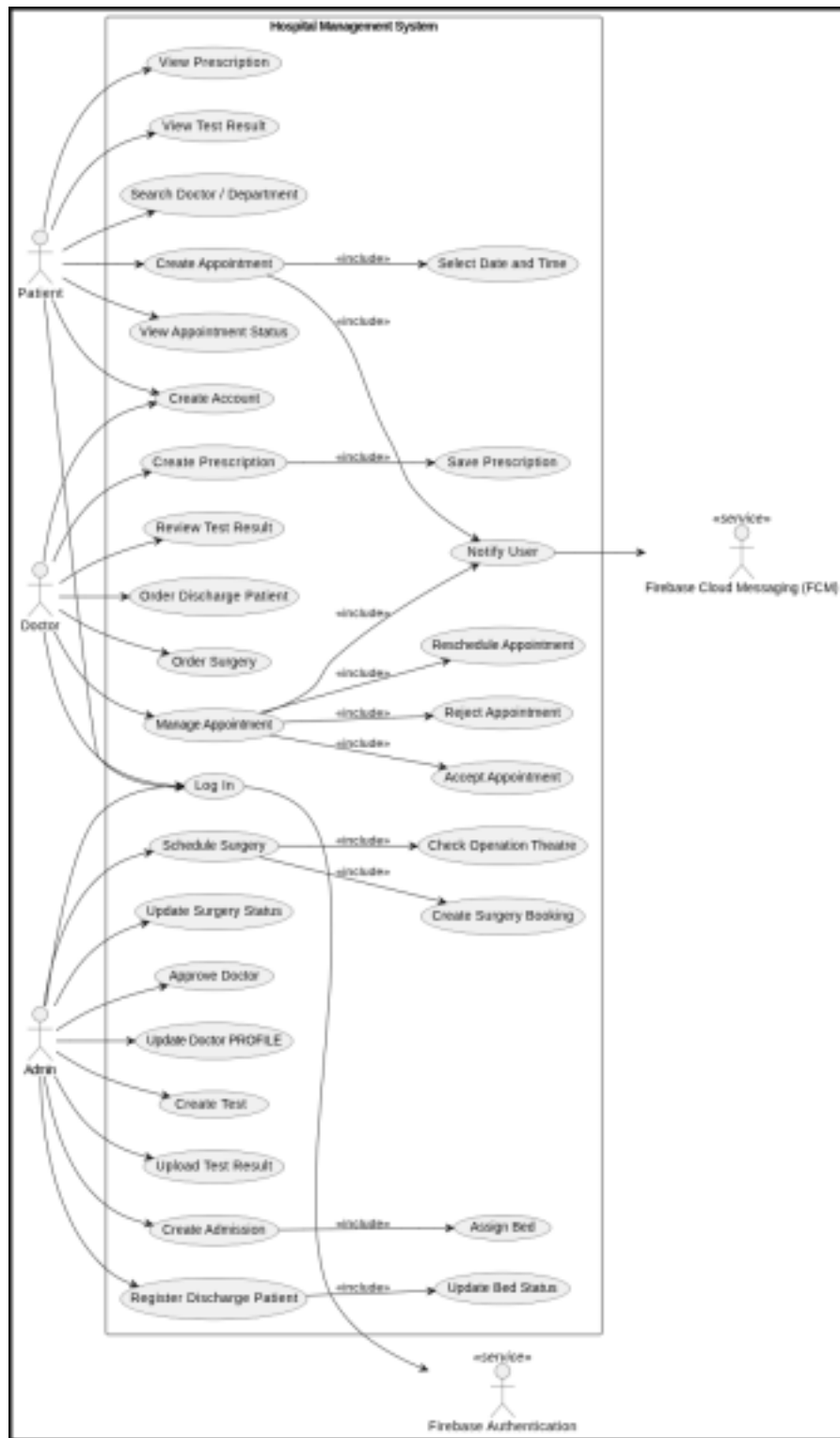


Fig: Use Case Diagram

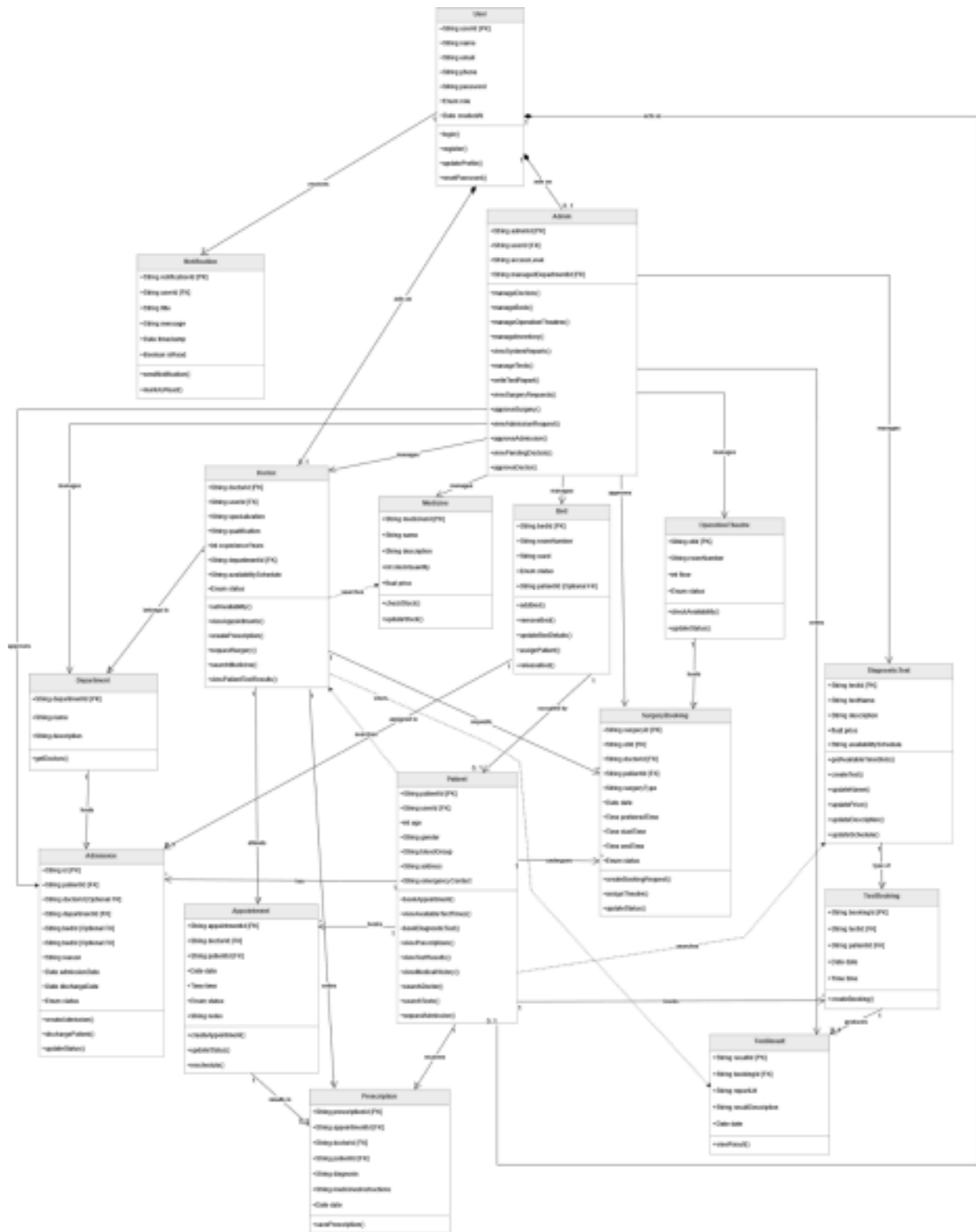


Fig: Class Diagram